

Erosion Control

Erosion control refers to the management of the working area to prevent soil particles being mobilised or entrained in water. The soil loss and resulting erosion risk determined specifies the extent of erosion control required on site. These requirements are outlined below.

Table A - Application of Calculated Erosion Risk (Based on Table 4.4.7 of IECA 2008)

Applicable Area ID	Erosion Risk Rating	Soil Loss Rate (t/ha/year)	Advance Land Clearing Allowed (weeks work)	Max days to Stabilisation	Staged Construction and Stabilisation of Earth Batters >6H:1V	Stockpiles stabilised
Disturbed Catchment	Very Low	0 to 150	8	30 (60%)		
	Low	150 to 225	8	30 (70%)		
	Moderate	225 to 500	6	20 (70%)	✓	
	High	500 to 1500	4	10 (75%)	✓	✓
	Extreme	> 1500	2	5 (80%)	✓	✓

Up to date weather forecasts (BOM) should be monitored prior to clearing areas. Clearing smaller areas is a preferred option .

Typically, priority should be given to retaining existing vegetative cover as long as possible before stripping, however it is acknowledged that, given the nature of the proposed works, this staging may not be possible. Instead, there will be an increased focus on sediment control.

Typically, until drainage and sediment controls can be installed all existing ground cover shall be retained. Clearing outside those areas reporting to sediment control shall be prohibited and may be prevented by installing flagging on the boundary. Retaining existing vegetative cover is particularly important in flow paths and drainage lines, and clearing of these areas should be postponed until construction works are scheduled to commence in these areas.

Where practical disturbed areas should be stabilised as soon as possible. Various applicable stabilisation measures are presented in **Table A**. Erosion control will be particularly important on batters and any area not directly flowing to the downstream sediment basins. Refer to Appendix A drawings for details.

Table 4 - Application of Erosion Control Measures to Soil Slopes (reproduced from Table 4.4.13 of IECA 2008)

Flat Land (<1 in 10)	Mild Slopes (1 in 10 – 1 in 4)	Steep Slopes (steeper than 1 in 4)
Erosion Control Blankets	Bonded Fibre matrix	Bonded Fibre Matrix
Gravelling	Compost Blankets	Cellular Confinement Systems
Mulching	Erosion Control Blankets, Mats and Mesh	Compost Blankets
Revegetation	Mulching well anchored	Erosion Control Blankets, Mats and Mesh
Rock Mulching	Revegetation	Revegetation
Soil Binder	Rock Mulching	Rock Armouring
Turfing	Turfing	Turfing
Coir logs	Coir logs	Coir logs

If temporary erosion control is required (i.e. due to work delays, or unexpected rain events) erosion control blankets (i.e. geofabric) or soil binder should be applied to exposed surfaces. Soil binders may also be utilised to further reduce the risk of erosion and help supplement watering for dust suppression.

Where practicable, it is recommended that mulch be utilised for erosion protection on completed areas. Mulch should be applied progressively as design levels are achieved. Alternatively, soil binder or hydromulch can be applied to stabilise completed areas, these products must be applied as per the manufactures specifications to achieve the desired erosion protection.